

FULLSTACK DEVELOPMENT

DATE:21.05.20026

CHAPTER:2

DAY:4

TOPIC:

Structure of Java Program

java syntax rules

Data Types in Java

Variables

Types of variables

Rules for variables

Structure of Java Program

A Java program contains different parts that help the program execute properly.

EXAMPLE:

```
puplic class helloworld{  
    public static voidmain(string[]args){  
System.out.print ln(“hello word”);}  
};
```

Java Syntax Rules

Java syntax rules are the basic rules that must be followed while writing a Java program. If the rules are not followed, the compiler shows errors.

- File name and class name must be same
- Java is case sensitive
- Every statement ends with ;
- Program starts from main()
- Keywords cannot be variable names
- Curly braces are mandatory
- Strings use double quotes " "

Data Types in Java

Data types are used to define what type of data a variable can store. Java contains two main types of data types.

- Primitive Data Types
- Non-Primitive Data Types

1. Primitive Data Types:

Primitive data types store simple values directly.

DATA TYPES:

Byte – 1 byte

Short – 2 byte

Int – 4 byte

Float – 8 byte

Long – 4 byte

Double-8 byte

Chr-2 byte

Boolean-1 byte

2. Non-Primitive Data Types

Non-primitive data types store references to objects.

Types

- String
- Array
- Class
- Interface

VARIABLES:

Variables are memory containers used to store values or information in a program. They are also called data holders.

SYNTAX:

Datatype variableName = value;

Example:

```
int age = 20;
```

TYPES OF VARIABLES:

Local Variable

A local variable is declared inside a method and can be accessed only within that method.

Instance Variable

An instance variable is declared inside a class and accessed through objects.

Static Variable

A static variable is declared using the `static` keyword and shared among all objects.

RULES FOR VARIABLES

- Variable names should begin with a letter, \$ or _.
- Variable names must not start with numbers.
- Spaces are not allowed in variable names.
- Java keywords such as `class`, `int`, and `static` cannot be used as variable names.

